AIML Summer Internship 2026 | IIHMF, MNNIT Allahabad

Employee Attrition Prediction System

Predicting employee attrition risk using classification models and deploying a live decision-support tool for HR teams

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Problem Statement

Organizations lose skilled employees every year, often without warning. Replacing them costs time, money, and institutional knowledge.

Goal: Using historical HR records, predict whether a given employee is likely to leave the organization — so HR can act before it happens, not after.

A binary classification problem: Attrition = Yes / No

1,470

employee records analyzed

16.1%

of employees left the company

Project Objectives

1. Clean & explore the HR dataset through thorough EDA
2. Engineer meaningful new features from raw HR fields
3. Balance the imbalanced target using SMOTE
4. Train & compare Logistic Regression, Decision Tree, Random Forest, XGBoost
5. Evaluate using Accuracy, Precision, Recall, F1, ROC-AUC
6. Deploy the best model as a live Streamlit web app

Dataset Description

IBM HR Analytics Employee Attrition Dataset

1,470	35	0	5.2 : 1
employee rows	attributes	missing values	class imbalance ratio

Key attribute groups:

- Demographics — Age, Gender, MaritalStatus, DistanceFromHome
- Compensation — MonthlyIncome, DailyRate, PercentSalaryHike, StockOptionLevel
- Role & Org — Department, JobRole, JobLevel, BusinessTravel
- Satisfaction — JobSatisfaction, EnvironmentSatisfaction, WorkLifeBalance
- Tenure — YearsAtCompany, TotalWorkingYears, YearsSinceLastPromotion

Methodology — End-to-End Pipeline



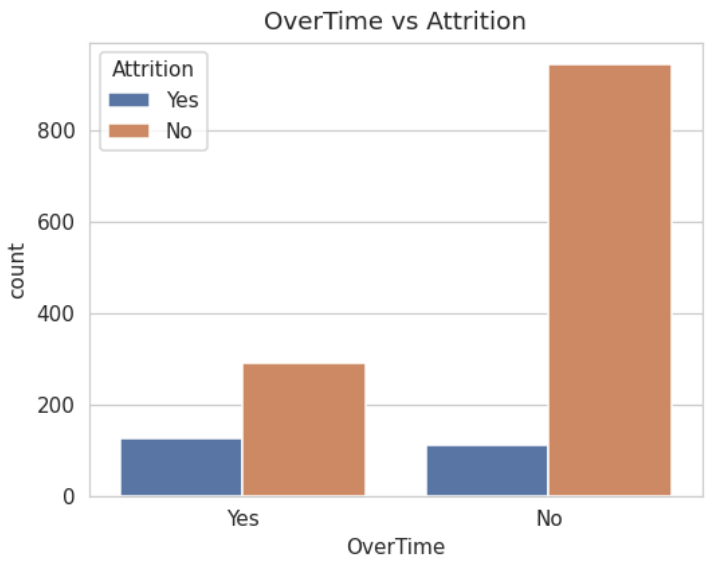
Follows the standard supervised ML lifecycle, ending in a deployed, user-facing prediction tool.

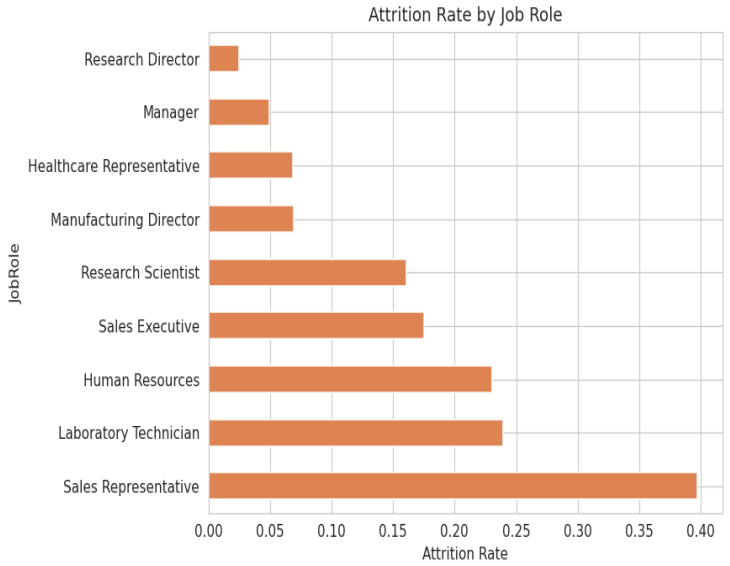
- Train/test split: 80% / 20%, stratified on Attrition
- SMOTE applied only on the training set to avoid data leakage
- Same preprocessing reused at inference time inside the Streamlit app

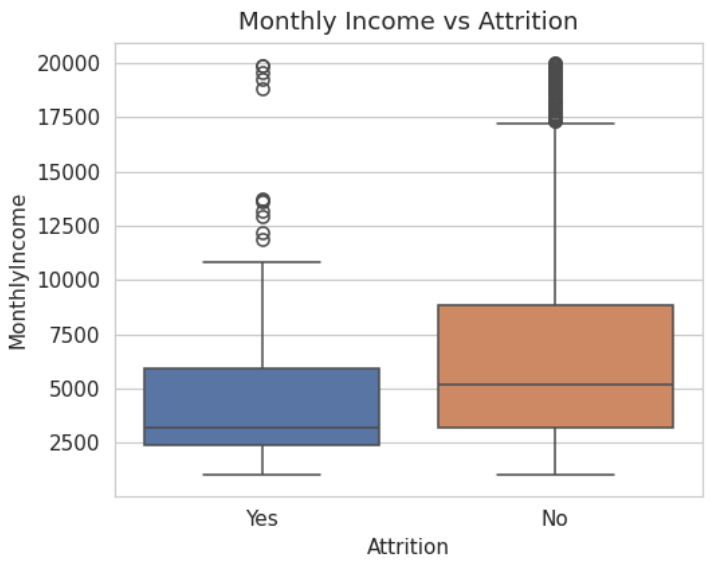
Data Preprocessing

✓ **Dropped 4 constant columns**	EmployeeCount, StandardHours, Over18, EmployeeNumber carry no predictive signal
✓ **Verified data quality**	Zero missing values, zero duplicate rows — no imputation needed
✓ **Encoded the target**	Attrition mapped to 1 (Yes) / 0 (No)
✓ **Label-encoded 7 categorical fields**	BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, OverTime
✓ **Scaled all features**	StandardScaler applied — required for Logistic Regression to converge well

Exploratory Data Analysis — Key Insights







- Employees who work OverTime leave at a much higher rate
- Sales Representatives & Lab Technicians show the highest attrition by role
- Leavers tend to have lower Monthly Income than those who stay

Feature Engineering

TenureRatio

YearsAtCompany / (TotalWorkingYears + 1)

IncomePerYearExp

MonthlyIncome / (TotalWorkingYears + 1)

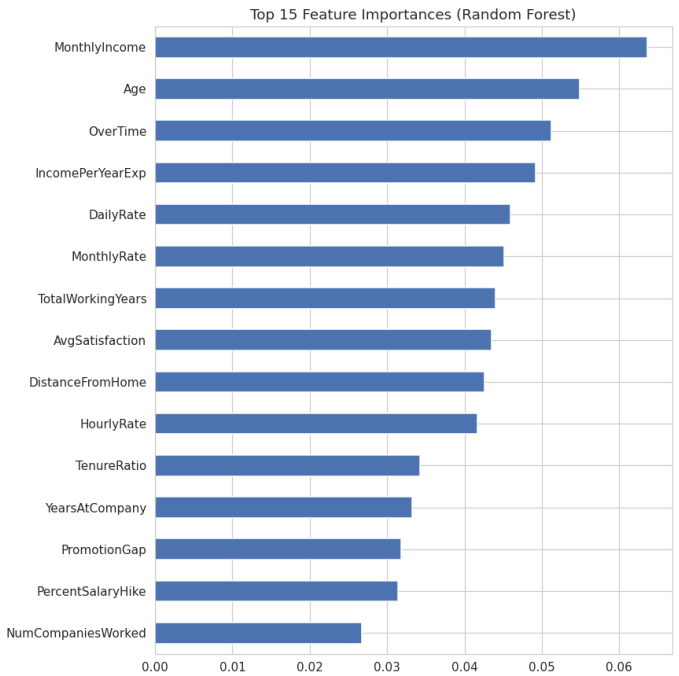
PromotionGap

YearsAtCompany – YearsSinceLastPromotion

AvgSatisfaction

Mean of 4 satisfaction & work-life scores

Top 15 Feature Importances (Random Forest)



2 of 4 engineered features rank in the top 10 by importance:

- IncomePerYearExp — rank #4
- AvgSatisfaction — rank #8

Models Used

Logistic Regression ✅

Linear, highly interpretable baseline. max_iter=1000. Deployed model.

Decision Tree

Single tree, depth capped at 6. Fast, fully interpretable.

Random Forest

Bagged decision trees, tuned via 3-fold GridSearchCV, class_weight='balanced'.

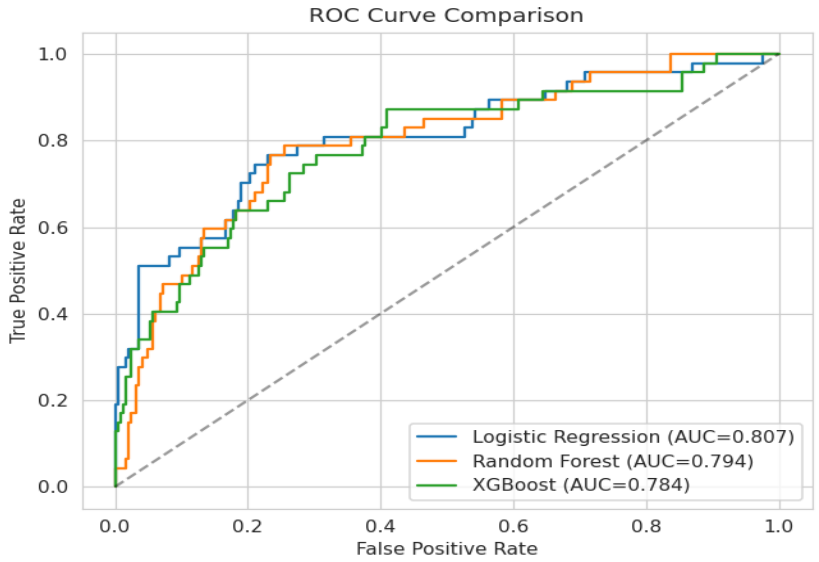
XGBoost

Sequential gradient boosting. 300 estimators, depth 5, lr=0.05.

Class imbalance (16% positive) handled via SMOTE oversampling on the training set before all four models were trained.

Results — Model Comparison

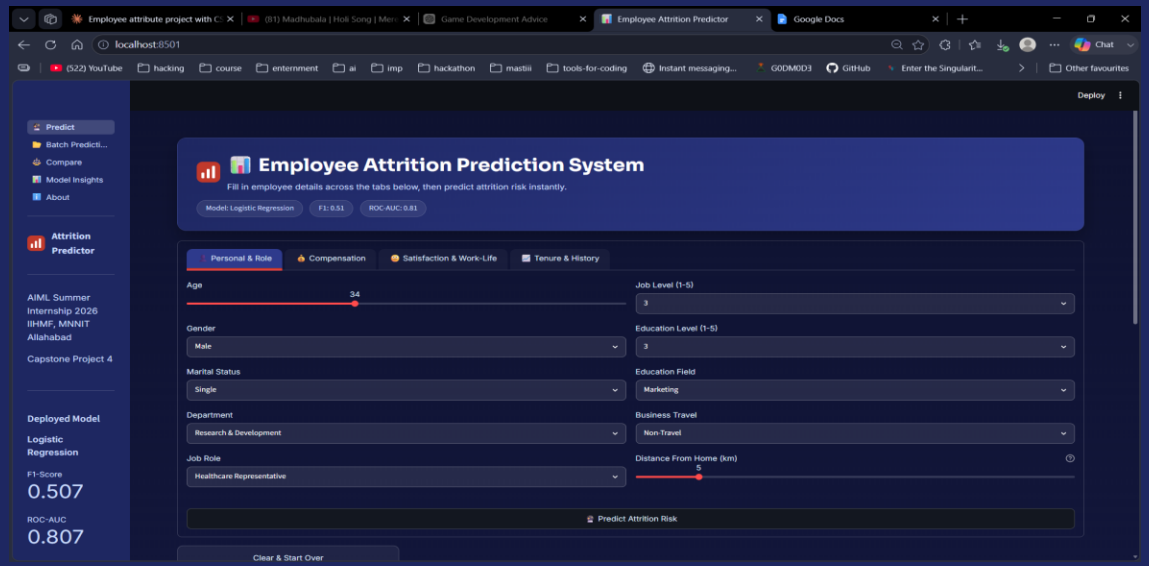
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	**0.762**	**0.379**	**0.766**	**0.507**	**0.807**
Decision Tree	0.810	0.435	0.638	0.517	0.736
Random Forest	0.847	0.538	0.298	0.384	0.794
XGBoost	0.864	0.640	0.340	0.444	0.784
LR (class_weight, no SMOTE)	0.752	0.362	0.723	0.482	0.813

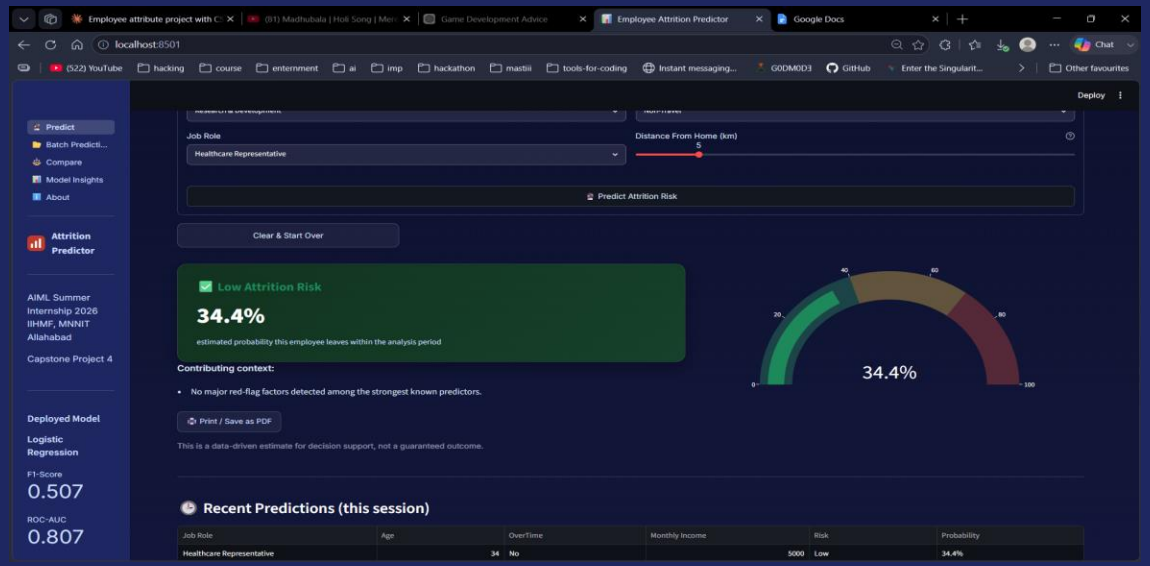


ROC shown for the original 3-model run — see table for all 5 variants

Decision Tree edges out Logistic Regression on F1 (0.517 vs 0.507) — but they're within 0.02, so treated as tied. Logistic Regression wins the tie-break on ROC-AUC (0.807 vs 0.736) and is deployed.

Live Demo — Streamlit App





Predict page — form (left) and live result with gauge (right). Also includes Batch Prediction, Compare, and Model Insights pages.

How it works:

- Fill in employee attributes in the form
- App re-creates the 4 engineered features
- Same scaler & encoders from training are applied
- Returns risk label + probability instantly

Conclusion

- Built a complete pipeline: preprocessing → EDA → feature engineering → SMOTE balancing → 4-model training → evaluation → deployment.
- OverTime, low relative income, frequent travel, and low satisfaction are the strongest attrition drivers.
- Logistic Regression was deployed via a defensible tie-break rule (F1 within 0.02 → best ROC-AUC wins) — not just the top scorer on one metric.
- Delivered as a 5-page Streamlit app (Predict, Batch Prediction, Compare, Model Insights, About), plus an optional custom FastAPI + HTML/CSS/JS deployment.

Future Scope

1. Add SHAP explainability so HR sees *why* an employee was flagged
2. Stack models to combine Logistic Regression's recall with tree-model precision
3. Collect longitudinal data for time-to-attrition survival modeling
4. Integrate into an HRIS with scheduled batch scoring & manager alerts
5. Audit for fairness across gender, age, and marital status